

Project Title

Recipe Finder

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

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Submitted by
Sreeya (2500032243)
Hasini (2500030238)
Pravalika (2520030402)

Team Number:
Section: 09

Under the guidance of

Mr. Vinay Kumar
Assistant Professor, CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

Recipe Finder is a DSA-based application designed to help users discover suitable recipes using the ingredients they already have. Instead of manually browsing through a large collection of recipes, users can enter available ingredients, and the system efficiently identifies relevant recipes. The project focuses on applying advanced Data Structures and Algorithms to solve real-world searching and optimization problems. String-matching techniques such as KMP and Rabin-Karp can be used for efficient ingredient and recipe-name searching, while dynamic programming can help select recipes based on ingredient availability and constraints. Hash-based data structures enable faster lookup, and optimization techniques can improve recipe recommendations. The system aims to provide quick and relevant results even when the recipe database is large. By combining practical recipe management with advanced DSA concepts, the project demonstrates how algorithmic techniques can be applied to develop an efficient, useful, and user-friendly solution while also helping reduce food wastage.

Signature of the Guide